

Homework NLP Omangali Hemamal 23B071564

$$x_1 = 0.35$$

$$x_2 = 0.9$$

$$w_{11} = 0.0991$$

$$w_{12} = 0.3971$$

$$w_{21} = 0.7176$$

$$w_{22} = 0.5928$$

$$w_{13} = 0.4749$$

$$w_{23} = 0.7525$$

$$x_1 \xrightarrow{w_{11}} h_1 \rightarrow y_3$$

$$x_2 \xrightarrow[w_{12}]{w_{21}} h_2 \xrightarrow{w_{23}} y_3$$

$$f(z) = \frac{1}{1 + e^{-z}}$$

$$(h_1)$$

$$z_1 = x_1 w_{11} + x_2 w_{21}$$

$$z_1 = (0.35)(0.0991) + (0.9)(0.7176)$$

$$z_1 = 0.034685 + 0.64585 = 0.75525$$

$$h_1 = f(z_1) = \frac{1}{1 + e^{-0.75525}} \approx 0.6800$$

$$(h_2)$$

$$z_2 = x_1 w_{12} + x_2 w_{22}$$

$$(x) z_2 = (0.35)(0.3971) + (0.9)(0.5928)$$

$$z_2 = 0.138985 + 0.53352 \approx 0.6725$$

$$h_2 = f(z_2) = \frac{1}{1 + e^{-0.6725}} \approx 0.6620$$

$$z_3 = h_1 w_{13} + h_2 w_{23}$$

$$z_3 = (0.6800)(0.4749) + (0.6620)(0.7525)$$

$$z_3 \approx 0.3229 + 0.4988 = 0.8217$$

$$y_3 = f(z_3) = \frac{1}{1 + e^{-0.8217}} \approx 0.6889$$

Выходной нейрон

ошибка:

$$y_+ = 0.5$$

$$\text{Error} = y_+ - y_3 = 0.5 - 0.6889 = -0.1889$$

Iteration 1.

$$(f') \quad f'(y) = y(1-y)$$

$$(y_+ =) y_+ = 0.5$$

$$\delta_3 = (y_+ - y_3) f'(y_3)$$

$$f'(y_3) = y_3(1-y_3) = 0.6889(1-0.6889) \approx 0.2148$$

↓ произв. активации нейрона.

$$\text{Error} = y_+ - y_3 = 0.5 - 0.6889 = -0.1889$$

$$\delta_3 = (-0.1889)(0.2148) \approx -0.0406$$

$$(\text{he}) \quad \delta_5 = f'(h_5) (w_{53} \delta_3)$$

$$f'(h_5) = 0.6800(1-0.6800) = 0.2176$$

$$w_{53} \delta_3 = 0.4749 (-0.0406) = -0.01928$$

$$\delta_5 = 0.2176 (-0.01928) \approx -0.00420$$

$$(h_2): \delta_2 = f'(h_2)(w_{23} \delta_3)$$

$$f'(h_2) = 0.6620(1 - 0.6620) = 0.2239$$

$$w_{23} \delta_3 = 0.7525(-0.0406) = -0.03055$$

$$\delta_2 = 0.2239(-0.03055) \approx -0.00684$$

$$\eta = 0.1$$

$$w_{13}: \Delta w_{13} = \eta \delta_3 h_1 = 0.1(-0.0406)(0.8700) = -0.0028$$

$$w_{13}^{\text{new}} = 0.4749 + \Delta w_{13} \approx 0.4721$$

$$\Delta w_{23} = 0.1(-0.0406)(0.6620) = -0.00269$$

$$w_{23}^{\text{new}} = 0.7526 - 0.00269 = 0.7499$$

↓
for h_3 .

$$\Delta w_{11} = 0.1(\delta_3 x_1) = 0.1(-0.00420)(0.35) = -0.000147$$

$$w_{11} = 0.09033 - 0.000147 = 0.08895$$

$$\Delta w_{21} = 0.1(-0.00420)(0.9) = -0.000378$$

$$w_{21}^{\text{new}} = 0.7546 - 0.000378 = 0.7127 \quad \text{for } h_2$$

$$w_{12} = 0.33(\delta_2 x_1) = 0.1(-0.00684)(0.35) = -0.000239$$

$$w_{12}^{\text{new}} = 0.3978 - 0.000239 = 0.39686$$

$$w_{22} = 0.1(\delta_2 x_2) = 0.1(-0.00684)(0.9) = -0.0006156$$

$$w_{22}^{new} = 0.5928 - 0.000686 = 0.5922$$

II Iteration

$$y_1 = (x_1 \cdot \Delta w_{11}) + (x_2 \cdot \Delta w_{21}) = (0.35 \cdot 0.099) + (0.9 + 0.2973) = 0.03965 + 0.7875 =$$

$$0.75237$$

$$f(y_1) = \frac{1}{1 + e^{-0.25237}} = 0.579552$$

$$y_2 = (x_1 \cdot \Delta w_{12}) + (x_2 \cdot \Delta w_{22}) = (0.36 \cdot 0.396882) + (0.9 - 0.593889) = 0.138887 + 0.5327009 = 0.57238$$

$$f(y_2) = 0.555855$$

$$y_3 = (h_1 \cdot \Delta w_{13}) + (h_2 \cdot \Delta w_{23}) = (0.75233 \cdot 0.2694) + (0.0715 \cdot 0.8704) = 0.20267 + 0.58956 = 0.7872$$

$$f(y_3) = 0.58723$$

$$\sum error: 0.5 - 0.88723 = -0.38723$$

back-propagation

$$\delta_3 = y_3(1 - y_3)(y_4 - y_3) = 0.8872(1 - 0.6872)(0.5 - 0.6872) = 0.6872 \cdot 0.3128 \cdot (-0.1872) = -0.04024$$

$$\delta_2 = y_2 (y - y_2) (\Delta w_{23} - \delta_3) = 0.6519 (1 - 0.6819) \cdot 0.8704 \cdot (-0.04024) = -0.00294$$

$$\delta_5 = y_5 (1 - y_5) (\Delta w_{13} - \delta_3) = 0.629812 \cdot (1 - 0.629812) \cdot 0.02694 \cdot (-0.04024) = 0.00235$$

$$v_{23}' = 0.1 \cdot \delta_3 \cdot y_2 + \Delta w_{23} = (0.1 \cdot (-0.0402) \cdot 0.6639) + 0.8704 = 0.86723$$

$$\delta v_{33}' = 0.1 \cdot \delta_3 \cdot y_3 + \Delta w_{13} = 0.1 \cdot (-0.0402) \cdot 0.62587 + 0.2594 = 0.2662$$

$$\delta w_{11}' = 0.1 \cdot \delta_1 \cdot x_1 + \Delta w_{11} = (0.1 \cdot (-0.00238) \cdot 0.35) + 0.099056 = 0.09893$$

$$\delta w_{21}' = 0.1 \cdot \delta_1 \cdot x_2 + \Delta w_{21} = (0.1 \cdot (-0.06238) \cdot 0.9) + 0.79738 = 0.7972$$

$$\delta v_{22}' = 0.1 \cdot \delta_2 \cdot x_2 + \Delta w_{22} = (0.1 \cdot (-0.00284) \cdot 0.9) + 0.593889 = 0.5848$$

$$\delta w_{12}' = 0.1 \cdot \delta_2 \cdot x_1 + \Delta w_{12} = (0.1 \cdot (-0.00784) \cdot 0.35) + 0.39682 = 0.3965$$

III Iteration.

$$y_3 = (x_1 \cdot \Delta w_{13}') + (x_2 \cdot \Delta w_{21}') = (0.35 \cdot 0.09893) +$$

$$(0.9 \cdot 0.7972) = 0.0345255 + 0.71748 = 0.7521$$

$$f(y_3) = \frac{1}{1 + e^{-0.722}} \approx 0.6285$$

$$y_2 = (x_1 \cdot \Delta W_{12}) + (x_2 \cdot \Delta W_{22}) = 10,35 \cdot 0,3985) \\ + (0,9 \cdot 0,5898) = 0,138 \cdot 275 + 0,52532 = \\ = 0,8635.$$

$$f(y_2) = \frac{1}{1 + e^{-0,5001}} \approx 0,6604$$

$$y_3 = (h_1 \cdot \Delta W_{13}) + (h_2 \cdot \Delta W_{23}) = \\ = (0,7529 \cdot 0,2654) + (0,8655 \cdot 0,88775) \\ = 0,2001338 + 0,577127 = 0,7755$$

$$f(y_3) = \frac{1}{1 + e^{-0,7773}} \approx 0,6851$$

$$\text{Error}_3: 0,15 - 0,6851 = -0,051.$$